

Hardware Upgrade Plan

Storage & compute decision · Re-measured Aug 14, 2026 · Prices verified live Aug 14, 2026

THE ONE-PARAGRAPH VERSION

Since the first report, **the disk emergency solved itself** — the boot drive went from 99% full to **68%, with 282 GB free**. That takes the \$595 SSD off the urgent list. What did *not* improve is memory: the box is at **95 of 119 GB with swap 87% consumed**, and on this machine the render engine and the language model draw from **one shared 128 GB pool**. Meanwhile two prices moved hard in the wrong direction. A second GB10 is no longer \$3,099 — the \$2,999 listings are **backordered** and the real, shippable price is **\$4,100–4,313**. And drives are up ~46% since September. Your instinct on both counts was right: **16 TB genuinely beats 12 TB** (by 21–27% per terabyte), and **waiting will cost you money, not save it**. One correction, though: Amazon is *not* your only option, and it is not your cheapest.

282 GB

Free on the boot drive — was 17 GB. **Disk emergency over**, the SSD buy is deferred.

87%

Of swap consumed, with memory at 95 of 119 GB. **This is now the only real constraint**.

\$25.56

Per TB for 16 TB recertified — vs **\$32.42/TB** for 12 TB. The 16 TB call is correct.

\$4,100+

Real shippable price for a second GB10. The \$2,999 listing exists but is backordered.

THREE ANSWERS, UP FRONT

A Storage — buy this now

\$3,307

UNAS Pro 8 (\$862) + 5× 16 TB Exos X18 manufacturer-recertified (\$489 ea) in RAID 6 → **48 TB usable**, your 15 TB lands at 31% full, survives *two* simultaneous drive failures, three bays still open. Buy the drives **with** the chassis — page 4 explains why waiting is the expensive choice.

B Compute — the second GB10

\$4,100–4,313

Still the right box, at a worse price. But **do not fuse the two into one 256 GB machine** — that mode is for models too big for one box, and it is *slower*, not faster. Run them as **two independent workers**: box 2 takes the language-model server, box 1 gets its memory back. Page 6.

C The corrections

3 things

Amazon is not the only seller — the UNAS is in stock direct from Ubiquiti. **The 10G module is unnecessary** — the Pro 8 has a native 10 GbE copper port, saving ~\$60. And **the ZFS expansion warning you were given does not apply** to this NAS. Page 5.

On the "gradual capacity" warning you were quoted: that describes ZFS RAIDZ expansion. The UNAS Pro line runs **BTRFS**, not ZFS — so the specific claim that you'd only get your sixth drive's space back "as data is rewritten" appears to be **the right fact about the wrong product**. The practical advice it led to was still sound (buy the drives up front), but the reasoning behind it did not hold. Page 5 has what Ubiquiti actually documents, and what I could and could not verify.

What changed since the first report

Re-measured on your systems Aug 14, 2026. The first report was written against Aug 13 data — one of its two emergencies has since resolved, and the other one got worse.

The scoreboard

ITEM	AUG 13 (V1)	AUG 14 (NOW)	WHAT IT MEANS FOR THE BUY LIST
Boot disk	853 of 916 GB — 99% full , 17 GB free, under a week of runway	588 of 916 GB — 68% full, 282 GB free	Emergency over. The \$595 external SSD drops off the urgent list — deferred, not cancelled.
Memory	111 of 119 GB wanted; the Aug 9 lockup	95 of 119 GB used · swap 41 of 47 GB used (87% consumed)	Worse. This is the only real constraint left, and it is exactly what the second GB10 buys.
Out-of-memory kills	1 in July → 6 in Aug 1–13, accelerating	Zero in the visible log window	The Aug 9 guardrails (earlyoom, up 5 days) are holding — but the log only reaches back ~2 days, so this is a thin all-clear.
Video output	1,271 files Aug 1–13	1,391 files Aug 1–14	Trend intact — ~120 renders/day, still far above July's 719/month.
Modal cloud spend	\$403 MTD on day 13	\$331 metered · \$294 billed MTD on day 14	Pacing \$630–710/month . Below the \$1,000 trigger, above the line that justifies owned hardware.
Storage on hand	NAS 88% / 74%	Media volume 9.7 of 11 TB (90%) · SSD volume 1.4 of 1.9 TB (73%)	~15 TB to migrate. This is now the live decision.

Why the disk freed up: the Aug 13 emergency sweep cleared roughly 265 GB — the model folder alone went from 398 GB to 262 GB. Worth being clear-eyed about: that was a **manual cleanup, not added capacity**. You now have **16× the headroom** you had a day ago — on the same net-fill rate that gave v1 "less than a week," that is **months, not days** — but it is the same wall, just further out.

The memory number is the one to read twice. Only 20% of memory and **13% of swap** remain free while a render runs at 93% GPU. On the GB10 the graphics processor and the system share **one 128 GB pool** — the render engine and the language-model server are drinking from the same well. That is precisely why adding a graphics card with its own 32 GB does not fix this, and why a second 128 GB box does. earlyoom's kill threshold sits near where swap is now: the guardrails are working, but they are working *hard*.

Still not done from v1's free list: log retention was never raised — the journal still only reaches back to Aug 12, which is why the "zero OOM kills" line above carries an asterisk rather than a victory lap. Also still uncollected: ~49 GB in the user cache folder, and the container image prune. Both remain **\$0**.

Settled: 16 TB, not 12 TB

You were right, and the margin is bigger than you'd expect. Every price below was pulled live from ServerPartDeals' catalogue on Aug 14, 2026, filtered to drives actually in stock.

Cost per terabyte — in-stock drives only

CAPACITY & DRIVE	PRICE	PER TB	CONDITION	VERDICT
12 TB Dell/Seagate Exos X16 ST12000NM005G	\$389	\$32.42	Refurb · 2 yr	The worst value on the board. This is the drive the old plan called for.
16 TB Toshiba MG08 MG08ACA16TE	\$409	\$25.56	Seller refurb · 2 yr	Cheapest per TB that will actually work. 21% better than the 12 TB.
16 TB Seagate Exos X18 ST16000NM000J	\$449	\$28.06	Seller refurb · 2 yr	Middle option; same drive as below with a weaker warranty.
16 TB Seagate Exos X18 ST16000NM000J	\$489	\$30.56	Manufacturer recert · 3 yr	The recommended drive. Factory-recertified, not seller-refurbished, and the longest warranty available.
18 TB Seagate Exos X18 ST18000NM002J	\$539	\$29.94	Refurb · CMR	No better per TB than 16 TB, and \$50 more per drive up front.

The headline: 12 TB → 16 TB drops cost per terabyte from **\$32.42 to \$25.56–30.56** — a **21–27% improvement** — while cutting bays, cables and failure points for the same capacity. Going further gains nothing: 18 TB is no cheaper per TB, and every in-stock 20 TB is a shingled drive (page 4). For contrast, a **new WD Red Pro 16 TB** is **\$704.99** at Newegg — **\$44.06/TB**, the price you're avoiding. **16 TB is the sweet spot, and it is not close.**

How many to buy, and at what redundancy

CONFIGURATION	DRIVES COST	ALL-IN W/ NAS	USABLE	\$/USABLE TB	FIT FOR YOUR 15 TB
5× Exos X18 · RAID 6 PICK	\$2,445	\$3,307	48 TB	\$68.90	31% full. Survives two failures. Three bays spare.
5× Exos X18 · RAID 5	\$2,445	\$3,307	64 TB	\$51.67	23% full, but one parity drive across 16 TB members — see the rebuild warning.
6× Toshiba MG08 · RAID 6 VALUE	\$2,454	\$3,316	64 TB	\$51.81	Same money as the pick, 16 TB more usable — trades the 3-year factory warranty for a 2-year seller one.
6× Exos X18 · RAID 6	\$2,934	\$3,796	64 TB	\$59.31	The buy-once-cry-once option if the budget stretches.

Why RAID 6 at this drive size. Ubiquiti documents rebuild at up to 200 MB/s — 11–12 hours for an 8 TB drive, so **roughly a full day for a 16 TB rebuild**, and that's the optimistic figure on an idle box. Under RAID 5 you spend that entire day with **zero** redundancy, on recertified drives, while every surviving drive is read end to end. RAID 6 costs one drive's capacity and deletes that whole class of bad night.

Where to buy — and when prices come down

You asked whether anywhere beats Amazon. Yes, substantially — but two of the cheapest listings you'll find are traps.

Sources, best first

1

ServerPartDeals — manufacturer-recertified enterprise drives

\$25.56–30.56

per TB · in stock now

Source of the \$489 Exos X18 (3-yr) and the \$409 Toshiba MG08. Datacenter drives returned, re-tested and re-warranted. Against a new WD Red Pro at \$705 you save **\$216–296 per drive — \$1,080–1,480 across five**, with a 30-day return window on top of the warranty.

 serverpartdeals.com — 10–20 TB enterprise drives

2

Ubiquiti direct — for the chassis, not the drives

\$862

surcharge included

The UNAS Pro 8 is **in stock at ui.com right now** — \$799, or **\$862 with the import surcharge**. That's the correction to "Amazon is the only place": it's available at the source, and Micro Center's tariff-inclusive price lands the same or higher. The 10G direct-attach cable is **\$13** from the same page.

 store.ui.com — UNAS Pro 8, in stock

3

Amazon / Newegg / Best Buy — the fallback, not the plan

\$36–44

New retail 16 TB runs **\$579–705** (\$36–44/TB) — worth it only if you want a fresh 5-year manufacturer warranty and will pay **40–70% more per terabyte** for it.

 [Newegg](https://www.newegg.com) — WD Red Pro 16 TB, \$704.99

Two traps that look like the best deals on the page.

① **SAS drives will not work.** Several of the cheapest 16 TB listings — a \$399 Exos X16, a \$549 Exos X18 — are **SAS 12Gb/s**. The UNAS Pro 8 takes **SATA**. Sorting a catalogue by price per terabyte walks you straight into these; check the interface on every line.

② **Every in-stock 20 TB is shingled.** Both 20 TB drives and the cheapest 18 TB (\$449) are **Host-Managed SMR** — overlapping tracks that require the host to schedule writes, which in a normal NAS array means catastrophic write performance and ugly rebuilds. They show the best \$/TB on the board. **Don't.**

When do drive prices come down — and why they're up

Three forces stacked at once, none resolving this year:

- **AI datacenters bought the supply.** WD and Seagate have confirmed their **entire 2026 production is sold out** to hyperscalers; the consumer channel gets the remainder.
- **Tariffs** on East Asian electronics add **10–13%** to landed cost — that's the surcharge line at UNAS checkout.
- **Flash got pulled away.** NAND output is being diverted to high-bandwidth memory for AI accelerators, pushing buyers back onto spinning disks.

Result: HDD prices are up **~46% since September 2025**, at two-year highs, and consensus is that relief doesn't arrive **until 2027 at the earliest**, when new capacity ramps.

So: buy now or wait? Buy now. Waiting six months doesn't get a discount — on every published forecast it gets the same drives at a higher price, or out of stock. The honest caveat: these are *forecasts*, and forecasts miss. But no scenario on the table has 2026 storage getting cheaper, and you're at 90% on the media volume today. Being early costs little; being late means paying more for something you'll buy anyway.

The UNAS Pro 8 — and the truth about expanding it later

Specs confirmed from Ubiquiti product data, Aug 14, 2026.

What you actually get for \$862

SPEC	UNAS PRO 8	WHY IT MATTERS HERE
Networking	(2) 10G SFP+ · (1) 10 GbE RJ45	Saves ~\$60. v1 budgeted a 10GBASE-T module because the 7-bay is SFP+ only. The Pro 8 has native copper 10 G — plug it straight in.
Bays	8× 2.5/3.5" hot-swap	Five drives fills it to 62% — three bays for growth or a spare.
Cache / CPU	2× M.2 NVMe slots · quad-core ARM Cortex-A57 @ 2.0 GHz	Optional read cache, not needed for a media archive. The CPU is fine for file serving — don't expect it to transcode.
RAID levels	RAID 5, RAID 6, RAID 10	RAID 6 is available — which is what page 3 recommends at 16 TB.
Power supply	Dual hot-swap 550 W bays — one module in the box	Redundancy is an extra \$159 . Budget it or accept a single point of failure.
Availability	In stock at ui.com	No need to pay a marketplace reseller.

Does it expand? Yes — but buy the drives up front anyway

Ubiquiti documents the procedure: **Settings** → **Control Plane** → **Storage**, select the pool, select the RAID group, add the drive. The documented constraints are real and worth knowing before you plan around them:

- **Only "Unassigned" drives can be added**, and the system must be **completely idle** — not syncing, not repairing, no other storage operation running.
- **All drives in a group must be the same size** — mixing capacities limits the group to the smallest and slowest member.
- **Expansion is slow.** At ~200 MB/s (11–12 hours per 8 TB), adding a 16 TB member is **roughly a day of idle-system time** on a box meant to be serving Plex.

About the warning you were given. You were told that after expansion the reported capacity wouldn't reflect the new drive, that existing data keeps its old layout, and that space returns only "gradually, as data is rewritten" — genuinely bad news for a write-once archive. **That is real, well-documented ZFS RAIDZ behaviour.** The catch: **the UNAS Pro line runs BTRFS, not ZFS** — Ubiquiti's ZFS work is a future enterprise direction, not this box. Their documentation describes expansion as a synchronisation measured in hours per terabyte, which is the signature of a full restripe: the kind that *does* hand back the whole drive. **Caveat:** their help article blocks automated access, so this rests on the documentation summary plus the filesystem identification — treat it as "very likely fine," not "confirmed."

Which makes the question moot — deliberately. Buy five drives now and you never run the expansion path: you land at 31% full with three empty bays, whose best uses are a **replacement** for a failed drive and a **cold spare**, not growth. Ubiquiti's own guidance is to install drives before setup.

Carry-over rules, unchanged: the old 3 TB WD Reds stay **out** of this array (mixed sizes cap the group; one has 33,000 hours on it). The Plex database stays on the GEEKOM's NVMe — media moves to the NAS, the database never runs over a network share. And budget the migration honestly: the UGREEN's 1 GbE uplink caps the copy at ~110 MB/s — **28–30 hours** for the 11 TB side.

The second GB10 — what "fused as one" actually buys you

The question you asked — and the honest answer is not the one the marketing implies.

First, the price moved against you. The \$3,099 from v1 is effectively gone — the \$2,999 CDW listing exists but is **backordered**. What ships is **\$4,100** (days) or **\$4,313** (Newegg, 2 TB). **Your \$4,500 figure is the right number to plan against** — the LPDDR5X shortage inflating RAM and SSDs is inflating these too.

What two linked GB10s can do

The hardware is genuinely built for it: each box carries an **NVIDIA ConnectX-7 SmartNIC at 200 Gb/s**, and two units link **directly, no switch**, over one 0.5 m QSFP56 cable — **256 GB of unified memory** across the pair, with NVIDIA claiming **Llama 3.1 405B in FP4**.

Measured reality, from an independent two-node cluster review

StorageReview, testing Dell / GIGABYTE / HP GB10 boxes in dual-node configuration

WORKLOAD	THROUGHPUT	WHAT IT TELLS YOU
GPT-OSS-120B, pipeline-parallel, batch 128	554.69 tok/s	A model that cannot fit in one 128 GB box, running well.
GPT-OSS-120B, tensor-parallel , batch 128	252.01 tok/s	2.2× slower. Splitting a layer across boxes means 160 cross-box exchanges per token on an 80-layer model.

The reviewers' framing: clustering exists because "a single 128 GB box cannot hold every model that matters," and the platform is "optimized for individual exploration and learning, not for serving traffic."

The honest answer to "what is the second box's purpose when fused as one": fusing buys **capacity, not speed** — it runs a model too large for 128 GB, but does *not* make existing work faster, and the wrong parallelism mode makes it **less than half** as fast. 200 Gb/s is enormous for a network and still **far slower than memory inside one box**. That gap is the whole story.

Which is why you should not fuse them

Look at the box right now: **ComfyUI rendering at 93% GPU**, a **model server holding ~51 GB permanently**, and **87% of swap consumed** because those two fight over one 128 GB pool. Nothing there is waiting on a 405-billion-parameter model.

APPROACH	WHAT HAPPENS	VERDICT
Two independent workers	Box 2 takes the model server and its 51 GB reservation; box 1 gets that memory back — +2 render slots immediately , and box 2's spare ~68 GB adds 2–3 more. Swap pressure ends.	Do this.
Fused into one 256 GB machine	You gain a 405B model you have no use for and lose what you need — two machines running two workloads. Rendering doesn't distribute regardless.	Wrong tool.

Not either/or forever: the cable is \$15 and the link is a configuration, not a weld — run them split for production, cluster them for a session if you ever want a 120B-class model, then go back. **Buy the second box for the split; treat fusion as a capability you own, not the reason for the purchase.** Day one, box 2 gets box 1's guardrails *before* production work, or it inherits the identical lockup.

GB10 vs 5090 at the same price — and the revised order

Both land near \$4,500 now. That makes the comparison cleaner than it was in v1.

	SECOND GB10 · ~\$4,100–4,313	RTX 5090 · ~\$4,500 + A BUILD
Memory available to a job	128 GB unified — holds the 51 GB model server <i>and</i> renders	32 GB — cannot hold the model server at all
Raw speed per render	1× (same as today, but a second one running)	2.75–4.7× per job — genuinely faster
Fixes the actual constraint	Yes — frees 51 GB on box 1	No — the memory fight continues on box 1
True cost to be useful	\$4,100 + \$15 cable	\$4,500 + tower + \$699 RAM + possible PSU
Power / heat / noise	~140–240 W, sits on a shelf, silent	575 W card alone, needs a real tower, real cooling, real noise
Setup risk	Clone the working stack	Linux reimage, PSU connector verification, no chassis today

Your own reasoning holds up: the 5090 is faster per render, but it arrives with no rig, needs memory you'd buy at crisis prices, and adds meaningful power, heat and noise to a home office. The GB10 attacks the constraint you actually have — **memory** — and does it silently. The 5090 stays a *trigger* buy: revisit if Modal crosses ~\$1,000/month or a used card lands under \$3,000.

The revised order

WHEN	ACTION	COST
Now	UNAS Pro 8 + 5× 16 TB Exos X18 recert + 10G DAC. Order the drives with the chassis — prices only rise from here.	\$3,320
Now	2× pure-sine UPS (CyberPower CP1500PFCLCD). Still unbought, still the cheapest catastrophic-loss insurance you can buy — and now you have a NAS full of media to protect.	\$460
This month	Second GB10, run split not fused. Language-model server moves to box 2; guardrails cloned on day one.	\$4,100
Deferred	External 4 TB USB4 SSD — no longer urgent at 68% disk. Add it to the next order rather than making a trip for it.	\$595
On trigger	RTX 5090 — only if Modal crosses ~\$1,000/month or a used card drops under \$3,000.	~\$4,500
\$0	Raise log retention · prune ~49 GB of cache · container image prune · trim the model-server reservation	\$0

The two decisions in front of you now — storage this week, compute this month

\$7,880

If it has to be one thing: buy the **storage**. The media volume is at 90%, drive prices are rising on every published forecast, and the NAS is in stock today. The GB10 is the bigger capability jump, but its price is already inflated and the memory pressure is currently being *held* — badly, but held — by the guardrails. Storage is the one where waiting has a visible, quantified cost.

Fine print — where every number came from

Measured on your hardware, Aug 14 2026

- **Disk:** 588 of 916 GB used (68%), 282 GB free, single internal NVMe. Model folder 262 GB (was 398 GB on Aug 13). User cache 49 GB. No external drive attached.
- **Memory:** 95 of 119 GB used; swap 41 of 47 GB (87% consumed); available 23 GB. Anonymous process memory is only 17.5 GB — the balance is GPU/driver allocation out of the shared pool, which is precisely the point. **Guardrails:** earlyoom active 5 days, zero kills in the visible journal — but that journal only reaches Aug 12, a ~2-day window.
- **Output:** 1,391 files with August timestamps as of Aug 14. June (110) and July (719) carried from the Aug 13 census — a timestamp recount now under-reports them, since files have been moved or swept.
- **Spend:** Modal August month-to-date — \$331.27 metered, \$294.20 billed after \$30 credits, queried directly from billing; deployed render apps are \$307 of the metered total. **Storage to migrate:** ~15 TB (media 9.7 of 11 TB, SSD 1.4 of 1.9 TB, plus GEEKOM Plex drives).
- **No UPS installed** — NUT is not present, so v1's power-protection gap is still fully open.

Prices — verified live Aug 14, 2026

ITEM	PRICE	SOURCE / NOTES
Seagate Exos X18 16 TB, manufacturer recertified, 3 yr	\$489	ServerPartDeals · live catalogue, in stock, SATA
Toshiba MG08 16 TB, seller refurbished, 2 yr	\$409	Cheapest in-stock SATA 16 TB · \$25.56/TB
Seagate Exos X16 12 TB, refurbished	\$389	The 12 TB comparison point · \$32.42/TB
WD Red Pro 16 TB new (WD161KFGX)	\$704.99	Newegg · the new-retail anchor, \$44.06/TB
Ubiquiti UNAS Pro 8	\$799 / \$862	store.ui.com · in stock; \$862 surcharge-inclusive. 2nd PSU \$159, 10G DAC from \$13
ASUS Ascent GX10 (GB10, 128 GB)	\$4,100–4,313	Newegg \$4,313 (2 TB) · \$4,100 shipping in days · CDW lists \$2,999 but backordered

Market claims and where they come from

- **HDD prices up ~46% since Sept 2025, two-year highs;** WD/Seagate 2026 production sold out to AI datacenters; tariffs 10–13%; NAND diverted to HBM; relief not before 2027. [WinBuzzer](#) · [The Register](#) · [DatacenterDisk](#)
- **Two-node GB10 performance** — 554.69 tok/s pipeline-parallel vs 252.01 tensor-parallel, GPT-OSS-120B at batch 128; built for exploration, not serving. [StorageReview dual-node review](#)
- **200 Gb/s ConnectX-7, direct QSFP56 link, 256 GB combined, Llama 3.1 405B FP4 claim.** [NADDOD](#) · [Corti](#)
- **UNAS expansion constraints** — unassigned drives only, idle system, uniform sizes per group, ~200 MB/s sync. [Ubiquiti Help Center](#) (summary only — article blocks automated retrieval)

What I could not verify. ① Ubiquiti's expansion article wouldn't load, so page 5's restripe conclusion rests on their documentation summary plus the filesystem identification — high confidence, not certainty. ② That BTRFS-not-ZFS identification comes from community and press reporting, not a Ubiquiti spec sheet. ③ Drive prices are live but perishable — the catalogue moves daily, and upward. ④ Micro Center's in-store price wasn't retrievable; the ui.com figure is the one I stand behind.

On the earlier text you flagged: you were right to question it. Its central claim — the gradual-capacity warning — is a real behaviour attached to the wrong filesystem, and it cited "ibm" for a claim about Ubiquiti hardware, which is not a source that would know. I can't tell you where it came from; I can tell you it shouldn't be treated as fact.